

Code				Description	Example
When	Action Occurence	Synchronous		The Intervention works synchrononus with the action occurence	A tool that detects potential bias in real time during data analysis and prompts the user to consider alternative approaches
		Asynchronous		The Intervention works asynchronous with the action occurence	A tool that enables users to visualize potential model bias towards a certain subgroup through clicking a certain button in the GUI after the model has been built
	DS Process	Preparation		Defining Needs, Data Gathering, Data Creation, Profiling, and Data Wrangling (including data exploration)	A Jupyter Notebook plugin that could automatically suggest privacy-preserving methods when sensitive data is being cleaned and prepared
		Analysis		Experimentation, Exploration, Modeling, Verification, and Interpretation (exploration here does not include data exploration)	Tools can assist data scientists by providing in-line guidance on statistical methods and algorithms that minimize bias and ensure fairness
		Deployment		Monitoring and Refinement	Mandatory ethical compliance checks that ensure models meet ethical standards before they are used in decision-making process
		Communication		Dissemination and Documentation to non-technical audiences	Interventions that help ensure that data visualizations and reports are transparent and do not mislead stakeholders about the implications of the data.
	Stages of Human Information Processing	Information Acquisition		Information acquisition refers to the acquisition and registration of multiple sources of information	A Jupyter Notebook plugin could alert users when the data they are importing has historically been prone to bias or when the data lacks representation from certain groups
		Information Analysis		Information analysis involves conscious perception and manipulation of processed and retrieved information in working memory	An embedded tool in Jupyter Notebook could analyze the algorithms being used and suggest modifications or alternative algorithms that are known to reduce bias
		Decision and Action Selection		The Decision and action selection stage is where decisions are reached based on the iterations of the previous two cognitive processes	An intervention could include a decision-support system that prompts users to consider ethical implications at critical decision points
		Action Implementation		Action implementation involves the implementation as a response or action consistent with the decision choice. This stage involves different levels of machine execution of the choice of action, and typically replaces the hand or voice of the human	interventions to help scientists think critically about the downstream consequence of their conclusions from the DS process
Where	In-Situ	---		Directly within the DS deployment environment	VsCode, Jupyter Notebook, Jupyter Colab
	Ex-Situ	---		Stand alone tools, systems, webpages that are independent from the DS deployment environment	Independent websites, softwares
How	Intrusiveness	Hidden and Ignorable		These interventions operate in the background with minimal or no notification to the user	An intervention in a Jupyter Notebook could silently monitor for the use of deprecated or non-compliant data processing methods, logging this use for later review without interrupting the user's workflow
		Hidden and Not Ignorable		These interventions operate in the background but take action without requiring user engagement, ensuring that essential tasks are performed	An automated tool that corrects spelling errors in variable names without notifying the user, ensuring consistency and accuracy in the code
		Visible and Ignorable		These interventions are apparent but do not force interaction	A sidebar in Jupyter Notebook that displays ethical guidelines or suggestions that users can choose to engage with or ignore at will during their work
		Visible and Not Ignorable		These interventions require user engagement to proceed	A popup that requires user action before certain types of data, such as that about sensitive or protected groups, can be processed
	Interactivity	Interactive GUI		These interventions involve active participation or input from the user, such as tools that require users to make selections, provide feedback, or make decisions based on the provided information	An interactive module in Jupyter Notebook could simulate different data handling scenarios, asking users to choose the best ethical approach and providing instant feedback on their choices
		Static Information		Static interventions do not allow for user input but provide critical information or warnings	A static report generated by a tool within Jupyter Notebook that assesses the ethical implications of a project's data usage, available for review at the user's discretion
	Programing Requirement	Coding Needed		These interventions require the user to write or modify code to use the intervention	A Python library that requires users to input specific commands or code snippets to check for data biases or ethical concerns
		Coding not Needed		These interventions are accessible to all users regardless of programming skills	A graphical interface within Jupyter Notebook that allows users to apply data privacy measures through simple dropdown menus and toggle switches
	Levels of automation	1	0. No automation from the intervention	The Intervention offers no assistance: the human must take all decisions and actions.	---
		2	1. The intervention offers low level of automation, it mainly relies human for decision making and excution	The Intervention offers a complete set of decision/action alternatives	---
		3		The Intervention narrows the selection down to a few	---
		4		The Intervention suggests one alternative	---
		5		The Intervention executes that suggestion if the human approves	---
		6		The Intervention allows the human a restricted time to veto before automatic execution	---
		7	2. The intervention offers high level of automation, it mainly relies on itself for decision making and excution, and only informs human when necessary	The Intervention executes automatically, then necessarily informs the human	---
		8		The Intervention informs the human only if asked	---
		9		The Intervention informs the human only if it, the computer, decides to	---
		10	3. The intervention is fully automated	The Intervention decides everything and acts autonomously, ignoring the human	---
Note	We look into the tools' information mainly from four sources: (1) abstract of its paper if this tool has one; (2) github page description and information if this tool has one; (3) descriptive information of this tool from Zijie's survey website ; (4) the demo video of this tool if it has one. Shall there will be any conflict, we rely on these four resources in a decending order.				
Iteration 1	Aug 23rd	1. We change the defore/during/after behavior in "Action Occurence" into Sychoronus and Asynchronous, due to the reason that the latter pair can better capture the characteristics of the behavior. Through the coding process, 3 authors start coding the first 5, and a3 authors identified this change would be better suited to replace the action occurrence. 2. We reform the 10 levels of automations into 4 levels: No automation, low level of automation, high level of automation, and fully automated. The logestic for this change is 10 levels are too many for characterizing level of automations. It would be more convenient and straightforward to just categorize the level of automation into 4 levels. In this four levels, 0 stands for "No automation from the Intervention"; 1 stands for "The intervention offers low level of automation, it mainly relies human for decision making and excution"; 2 stands for "The intervention offers high level of automation, it mainly relies on itself for decision making and excution, and only informs human when necessary"; and 3 stands for "The intervention is fully automated".			Inter-rator reliability (1st Iteration): 31%
Iteration 2	Aug 30th	1. We clarify the coding needed definition 2. In-situ definition: unless we are forced to go outside of the DS coding environment, this will be IN-SITU. A tool can be both in-situ and ex-situ at the same time if it provide both functionalities. 3. Synchronous means a tool is always-on and consitantly providing help during the behavior, async means the intervention reacts and provides feedack after or not at the same time of excuting the behavior, it happens when the user expecitlly asks for it. 4. Data exploration is Preparation, everything in analysis is after data pre-processing. 5. If a tool explicitly tells us or hav a functionality that directly focus on a DS phase e.g., preparation, we should check that box. But if the tool only indirlectly imply a DS phase, we should not check that box. 6. Action implementation based on definition is "The final stage of action implementation refers to the actual execution of the action choice." Intervention examples deployed in a dashboard setting would come under "action implementation" for stages of human information processing, because dashboards are used by policymakers to take implementable actions. For tools which are designed to intervene/help data scientists who do not necessarily take the final decisions, most of such tools will not come under action implementation			Inter-rator reliability (2rd Iteration): 39%
Iteration 3	Sep 4th	1. Information acqasition indicates the process that the users absorbe infos gained by using the intervention tool, towards their decision-making in the DS model building process. 2. During the coding process, we should focus on the most important part of the functionality the intervention provides, versus trying to cover every aspect that tool provide, even though it only touches a little bit of that aspect. 3. Information analysis requires some level of explanability/interpretability, in the sense that the information analysis should be supported by some logestics, and for those black box intervention tools, if they don't help users to understand or build their own coganitive sense-making for a decision, it should not support information analysis. 4. The level of automation should focus on the implementation level, not the conceptual/theoratical way. If the functionality of the automation in the intervention provides behavior change then it can be stages 1 - 3. But if the user has full control in making behavior-based decisions then the intervention has no automation.			Inter-rator reliability (3rd Iteration): 72%
Iteration 4	Sep 5th	---			Inter-rator reliability (4th Iteration): 89%